

Image Analysis Report

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Executive Summary

Visual inspection reveals severe localized corrosion concentrated along a horizontal seam or weld line, indicative of significant metal loss and structural degradation. The stated severity is high, characterized by deep pitting and heavy scale formation that requires immediate NDT assessment and remediation.

Key Findings

#	Finding
1	Severe localized corrosion and heavy scaling at the central horizontal intersection point.
2	Visible deep pitting within the primary corrosion site; precise pit depth cannot be determined from the image alone.
3	Widespread atmospheric corrosion (rust blooming) across approximately 60% of the visible surface.
4	Horizontal seam or weld line shows preferential attack, suggesting potential galvanic or stress-related corrosion factors.
5	Vertical rust staining indicates active corrosion progression and moisture transport across the surface.
6	Complete coating failure with delamination and under-film corrosion spreading from the primary defect zone.

Localized detections (VLM, approximate)

- Severe Localized Pitting (conf. 92%): x=0.340, y=0.170, w=0.330, h=0.500
- Seam Corrosion (conf. 88%): x=0.000, y=0.460, w=1.000, h=0.080
- General Surface Corrosion (conf. 85%): x=0.010, y=0.230, w=0.220, h=0.650

Recommendations

#	Recommendation
1	Perform Ultrasonic Testing (UT) spot checks every 50mm along the horizontal line to quantify remaining wall thickness.
2	Conduct abrasive grit blasting to Sa 2.5 (near-white metal) standard to remove all corrosion products and reveal the base metal condition.
3	Perform Magnetic Particle Inspection (MPI) or Dye Penetrant Inspection (DPI) post-blasting to check for surface-breaking cracks in the seam.
4	Apply a high-performance multi-coat epoxy system with a zinc-rich primer for long-term atmospheric protection.
5	Evaluate the need for structural reinforcement or a patch plate repair if wall loss exceeds the design corrosion allowance (e.g., >20% loss).
6	Establish a 1-meter exclusion zone around the component if it is a pressurized vessel or piping until structural integrity is confirmed.